



AMAN DEMROT



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GitHub Profile



LinkedIn Profile

Malaviya National Institute of Technology, Jaipur | Roll No: 2023UCH1213

EDUCATION

Malaviya National Institute of Technology, Jaipur

2023 – 2027

B.Tech in Chemical Engineering

RBSE Board (Class XII)

2022

Engineers Point, Alwar

Percentage: 83.5%

RBSE Board (Class X)

2020

SSDM School, Alwar

Percentage: 83.6%

EXPERIENCE

Web Development Intern

May 2026 – July 2026

Bridgestone Technologies

Remote

- Assisted in building lightweight backend services with Node.js and Express, implementing RESTful API endpoints for dynamic user data.
- Designed and developed responsive user interfaces and dashboard components using **React.js** and **CSS**, ensuring seamless integration with backend APIs for dynamic data rendering.
- Optimized web application performance by streamlining database queries and state management, resolving minor bottlenecks to improve overall page load times and user experience.
- Optimized site performance and UI elements, leading to faster load times and improved user retention rates.

PERSONAL PROJECTS

IntelliDocs – RAG-Based AI Knowledge Assistant | [GitHub](#)

Python, FastAPI, LangChain, ChromaDB, Gemini 3.5 Flash

- Engineered a modular Retrieval-Augmented Generation (RAG) pipeline utilizing FastAPI and LangChain to execute semantic context searches over unstructured PDF data.
- Integrated Google's **gemini-embedding-001** and **gemini-3.5-flash** models to translate text into high-dimensional coordinates, producing context-grounded, cited answers.
- Designed a custom obsidian-themed Streamlit interface featuring dynamic text search and metadata-extracted source citations, deploying both client and backend servers asynchronously across Render and Streamlit Cloud.

PlacementTutor – Mentorship Booking API | [GitHub](#)

Node.js, Express.js, MongoDB, Mongoose, JWT

- Developed a scalable RESTful API using Express.js and Mongoose to manage multi-role user dashboards, sessions, and session bookings.
- Designed an atomic reservation engine utilizing MongoDB's **findOneAndUpdate** operator, fully resolving concurrent double-booking race conditions at the database level.
- Implemented a payment-protected checkout slot-lock system with a 5-minute timeout window using an efficient, lazy-cleanup evaluation pattern.
- Built a dynamic slot-splitting algorithm to partition larger mentor availability blocks into customized 15, 30, or 60-minute atomic segments.

TECHNICAL SKILLS AND INTERESTS

Languages: Python, C++ Tools: Excel, Power BI, Git, GitHub

Data & Analytics: Pandas, NumPy, SQL Soft Skills: Problem-solving, Communication, Leadership

Areas of Interest: Backend Development, js, Front-end Engineering, Product Growth

POSITIONS OF RESPONSIBILITY

Publicity Executive, Blitzschlag (Annual Cultural Fest)

2025

MNIT Jaipur

- Conducted inter-college outreach and promotional tours, engaging with diverse student communities to maximize festival visibility and registration.

Technical Executive, Sphinx (Technical Fest)

2025

MNIT Jaipur

- Facilitated technical workshops and outreach initiatives, interacting with technology enthusiasts to drive participation in technical competitions.

ACHIEVEMENTS

Best Filo Teacher

October 2025

Filo Education

- Mentored a large student base with top-tier instructor ratings.

First Prize – College Science Club Event

2024

MNIT Jaipur

- Secured first place in a highly competitive scientific contest.